

Information-Theoretic Characterization of Aging Information and Noise in On-Chip Slack Sensors

Work Summary and Contribution Relative to Prior Work

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Work Summary — Information-Theoretic Characterization of Aging Information and Noise in On-Chip Slack Sensors

This is the entry-point summary of the final project (TF) for **Teoria da Informação (TIP7266, PGETI/UFC, 2026.1)**. It records *what was used* — the data, the course

concepts and how each was applied, the results, and the deliverables.

1. What the project is

It places an **information-theoretic measurement layer** on top of an existing low-cost accelerated-aging burn-in platform and its synchronized (slack, T , V , time) logs. Treating the on-chip slack sensor as a **noisy communication channel** (latent aging = transmitter; reversible PVT + quantization = noise; RUL estimator = receiver), it re-expresses the noise, the PVT correction, and the recoverability of aging information in the language of entropy, mutual information, divergence, channel capacity and estimation. No new silicon, sensor or oven — the contribution is analytical and methodological.

Central upgrade over prior work: replace the linear coefficient of determination R^2 — which captures only linear dependence and yields no transferable unit — with **mutual information in bits**, valid where the physics (Arrhenius in T , super-linear in V) is non-linear.

2. Data used

Dataset	Device	Duration	Role
Tentative_1_20260520175A (aging device), DUT $\approx 125^\circ\text{C}$ / 1.40 V, PID	Device A	214.6 h , 768,884 post-soak samples @ 1 Hz	Entropy, MI, PVT correction, capacity, BSC, RUL, ICA
TCC vs SBCCI/SBCCI_TCC_...cs test circuit), non-PID bench	Device B (separate test circuit), non-PID bench	≈ 24 h stabilized, 2 s	KL bench comparison (noisy bench)
TCC vs SBCCI/TCC_SBCCI_...cs reference bench	Device B , PID reference bench	≈ 24 h stabilized, 1 Hz	KL bench comparison (reference)

Device A and device B are never mixed. All entropy/MI/correction/capacity/RUL/ICA results are device A; only the KL/JS bench-fidelity comparison is device B (where the two benches share the device, so the divergence isolates the bench/environment, not device-to-device variation).

3. Course concepts used, and how each was applied

Organized by the disciplina's *núcleos de conteúdo*. Every entry produced a concrete, data-backed number (§4).

Núcleo 1 — Principles & limits of communication

- **Communication-channel model of the sensor** — the unifying frame $S = S_{aged}(t) - \Delta_{PVT}(T, V) + \varepsilon_q$.

- **Source-coding theorem (limit only)** — used as the bound “any code $\geq H(S)$ ”; no codec implemented.
- **Channel capacity (Shannon)** — converts post-correction noise into an effective number of resolvable aging states.

Núcleo 2 — Measures of information (the core)

- **Entropy $H(S)$** — total uncertainty budget of the sensor output.
- **Joint & conditional entropy $H(S | T, V)$** — residual uncertainty once the environment is known.
- **Mutual information $I(S; T, V)$ (centerpiece)** — non-linear contamination measure replacing R^2 ; estimated by binned + Miller-Madow **and** KSG (Kraskov), cross-checked.
- **MI as an optimization objective** — “best burn-in” = minimize $I(S; T, V)$.

Núcleo 3 — Transmission, denoise, divergence, capacity

- **DeNoise** — the PVT correction *is* denoising: strip reversible $\Delta_{PVT}(T, V)$ to expose latent aging.
- **Kullback-Leibler divergence** — distance between the SBCCI and PID bench distributions; demonstrates KL’s asymmetry (not a metric).
- **Jensen-Shannon divergence** — symmetric, bounded “distance between validation environments.”
- **Binary Symmetric Channel (BSC)** — the binary aging-alarm decision; capacity $C_{BSC} = 1 - H_2(p)$, mapping to the failure-detection/prevention goal.
- **Noise / Gaussianity** — Gaussian closed-form KL vs empirical reveals non-Gaussian (heavy-tailed) excursions.

Núcleo 4 — Complexity, estimation, ICA

- **Description-length model selection (MDL/BIC), grounded in Kolmogorov complexity** — chooses linear vs non-linear PVT correction on the reversible residual using n_{eff} ; the paper’s theory section now derives the BIC-style penalty as the computable two-part-code stand-in for $K(x) = \min_{p: U(p)=x} l(p)$ and the invariance theorem, making the MDL verdict an explicit instance of Occam’s razor rather than an unexplained formula.
- **Least squares (OLS / GLS)** — linear PVT correction.
- **Maximum likelihood (MLE)** — non-linear (Arrhenius / super-linear) correction.
- **Method of moments** — descriptive second-order baseline.
- **Bayesian estimation** — Student- t posterior on the aging slope (Jeffreys prior, thinned to n_{eff}) $\rightarrow$ credible error bars on the RUL bound.
- **Cramér-Rao bound** — frequentist lower bound on the slope-estimate variance $\rightarrow$ detectability.
- **ICA + negentropy + blind source separation** — model-free recovery of the aging source from (S, T, V) .
- **Independence / non-Gaussianity measures** — excess kurtosis as the ICA contrast; inter-component MI as the separation score.

Supporting (statistical foundations, throughout)

Probability & random variables; **expectation, variance, correlation**; empirical **distributions; preprocessing / normalization; autocorrelation handling** → effective sample size n_{eff} ; **inference** (block-bootstrap confidence intervals, **surrogate null tests**).

Deliberately NOT used (scope decision: only what strongly correlates with the research)

- **Concrete source-coding codecs** (Huffman, Lempel-Ziv-Welch) — redundant with the entropy/autocorrelation results.
 - **Channel error-correcting codes** — no application here.
 - **Big Data / streaming tooling** (MapReduce/Spark) — engineering, not science, for this problem.
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4. Key results (all data-backed, auto-generated into the paper)

Device A — the information spine - Effective sample size: $n_{eff} = 1,224$ ($\tau_{int} = 314$ s), despite 768,884 raw samples — the honesty linchpin. - $H(S) = 3.43$ bits (27 levels); $H(S | T, V) = 2.30$ bits. - $I(S; T, V) = 1.13$ bits binned (KSG 1.22; 95% CI [1.09, 1.18]); $I(S; T) = 1.09$, $I(S; V) = 0.11$. - **Non-linearity result**: measured MI exceeds the Gaussian-equivalent of the linear $R^2 = 0.738$ (0.97 bits) by **26 %**. - **Surrogate null floor** (circular-shift): $p_{95} = 0.097$ bits → measured MI is **12× the floor** → real, not bias. - **PVT correction**: reversible coupling $0.656 \rightarrow 0.046$ bits = **93 % removed**; residual $\sigma = 0.84$ counts (prior PID: 0.73 — independent cross-check); **MDL selects linear**. - **Aging recovery**: stripping PVT strengthens $\text{corr}(S, t)$ from **-0.35** → **-0.73** — the latent aging trajectory is *exposed*, not just cleaned. - **Capacity**: SNR 2.36 → $C = 0.87$ bits → **1.8 distinguishable aging states**. - **BSC alarm**: 1 h $\approx$ coin flip ($C_{BSC} \approx 0$); **24 h → C_BSC = 0.99 bits**; full campaign $\approx$ certain. - **RUL (Cramér-Rao)**: min detectable rate **1.2 m-cnt/h**; min observation time **34 h** ($\ll$ 214 h campaign; $>$ the prior 23 h runs — explaining why those couldn't separate aging). - **RUL (Bayesian)**: slope posterior **-18.2 m-cnt/h**, **95% CrI [-19.2, -17.2]**; $P(\text{aging real}) \approx 100\%$; min-obs-time CrI [33, 36] h. Frequentist and Bayesian agree.

Device B — bench fidelity (KL) - Slack noise: SBCCI 3.23 vs PID 1.46 counts (2.2×), tracking temperature noise (2.64 vs 1.25 °C) → contamination is thermally driven. - **KL(SBCCI||PID) = 1.94 bits**, **KL(PID||SBCCI) = 1.01 bits** (asymmetric); **JS = 0.279 bits**. - Empirical KL (1.94) $>$ Gaussian closed-form (1.67) → non-Gaussian heavy tails.

ICA blind separation (device A, exploratory) - Model-free ICA recovers an aging component matching the regression S'_{corr} at **|corr| = 0.834** — independent corroboration of the PVT correction, using none of the measured T, V coupling. - Recovered sources strongly non-Gaussian (excess kurtosis +5.8 / +29.9). Separation **partial** (residual inter-IC MI 0.57 bits) — reported honestly, since the non-stationary trend + heavy tails only approximately meet ICA's assumptions.

5. Deliverables (file tree)

tf_teoría-da-informação/	
├── 00_Work_Summary.md	← this file
├── 01_Project_Overview.md	proposal / contribution / caveats
├── 02_Methodology.md	5-stage arc (acquire→transform→quantify→preserve→infer)
├── 03_Course_Topic_Mapping.md	concept→application→result (rubric evidence; selective)
├── 04_Dataset.md	empirical characterization of all three logs
├── 05_Execution_Plan.md	prioritized plan + status (all core items ✓)
├── analysis/	
│ ├── tier1_pipeline.py	entropy, MI, surrogate floor, PVT correction, capacity, BSC, CRLB + Bayesian RUL
│ ├── kl_benches.py	KL / JS divergence between benches
│ ├── ica_separation.py	FastICA blind separation
│ ├── make_figures.py	generates the 5 paper figures (PDF)
│ ├── results.json	machine-readable scalars
│ └── README.md	results tables + how to run
├── latex/	final document (compiles to a 12-page PDF)
│ ├── main.tex, notation.tex, references.bib	
│ ├── generated/values.tex	ALL scalars as macros (auto-filled by the scripts)
│ ├── sections/00_abstract ... 08_conclusion	
│ ├── tables/	dataset_summary, mi_vs_r2
└── figures/	5 figures (auto-generated by make_figures.py)

Reproducibility contract: numbers are never typed into prose. The scripts emit every scalar into latex/generated/values.tex as \newcommand macros and every figure into latex/figures/; re-running them on a new log regenerates the paper with zero transcription. Build: `cd latex && latexmk -pdf main.tex`. Run analyses: `cd analysis && python3 tier1_pipeline.py && python3 kl_benches.py && python3 ica_separation.py && python3 make_figures.py` (needs numpy, scipy, matplotlib).

6. Honesty caveats (carried in the paper)

- **Single aging device, single regime** — RUL is a *detectability bound and methodology*, not a validated multi-device lifetime prediction.
- $n_{eff} \ll n$ — the 1 Hz series is heavily autocorrelated; all confidence/credible statements use n_{eff} , with block-bootstrap intervals and a surrogate null floor.
- **Capacity** is used as an effective-resolution heuristic, not a literal optimal-coding claim.
- **ICA** is corroborative, not a clean unmixing.
- **The KL comparison is a different device** than the aging analysis; the two are never mixed.

Objectives, Applied Techniques, and Contribution Relative to Prior Work

Companion note to 00_Work_Summary.md, written to answer directly: *what is this project trying to do, what did each course technique achieve, and how exactly does it improve on each of the three prior papers in artigos/?*

1. Objectives

Place an **information-theoretic measurement layer** on top of an existing, already-built accelerated-aging burn-in platform and its (slack, T , V , time) logs — no new silicon, sensor, or oven. The specific goals:

1. Replace the prior work's linear coefficient of determination (R^2) with **mutual information** as the correct measure of environment→sensor contamination, valid where the physics (Arrhenius in T , super-linear in V) is non-linear.
2. Quantify, in bits, how much uncertainty is removed once temperature and voltage are logged (an entropy account of PVT correction).
3. Convert the sensor's post-correction noise into an effective **resolution** (distinguishable aging states) and an **alarm reliability** figure.
4. Derive a noise-limited **detectability bound** for remaining-useful-life (RUL) estimation — a bound and a methodology, not a lifetime prediction.
5. Do all of this with a statistically honest treatment of a short, heavily autocorrelated dataset (effective sample size n_{eff} , not raw sample count).

The unifying idea: treat the slack sensor as a **noisy communication channel** — latent aging is the transmitter, reversible PVT + quantization is channel noise, the RUL estimator is the receiver — and let entropy, mutual information, and capacity do the work that variance-based metrics cannot.

2. Techniques applied, and what each achieved

Course technique	Applied to	Result
Entropy $H(S)$	Total uncertainty of the raw slack reading	$H(S) = 3.43$ bits over 27 discrete slack levels — the uncertainty budget everything else is carved out of
Conditional entropy / mutual information $I(S; T, V)$	Environment-sensor coupling (replaces R^2)	1.13 bits binned (KSG cross-check 1.22); exceeds the Gaussian-equivalent of the prior linear $R^2 = 0.738$ by 26% — direct proof of non-linear coupling R^2 structurally cannot see

Course technique	Applied to	Result
KL / Jensen-Shannon divergence	Distributional gap between the SBCCI and PID validation benches	$D_{KL}(\text{SBCCI} \parallel \text{PID}) = 1.94$ bits (asymmetric, illustrating KL is not a metric); $JS = 0.279$ bits — a single transferable “distance between validation environments”
Channel capacity (Gaussian heuristic) + Binary Symmetric Channel	Post-correction noise floor; the binary aging-alarm decision	SNR 2.36 $\Rightarrow C = 0.87$ bits $\Rightarrow$ 1.8 distinguishable aging states ; alarm reliability climbs from a coin flip (1 h) to $C_{BSC} = 0.99$ bits (24 h)
MDL, grounded in Kolmogorov complexity / Occam’s razor	Choosing linear vs. non-linear PVT correction	Selects the linear model — reframed not as “the physics is linear” but as “not enough curvature information yet at this single stress point to pay for the extra parameters”
Cramér-Rao bound + Bayesian estimation	RUL as estimation of a latent aging slope	Min detectable rate 1.2 m-cnt/h , min observation time $\approx$ 34 h (frequentist); Bayesian posterior agrees (slope —18.2 m-cnt/h, 95% CrI, $P(\text{aging real}) \approx 100\%$)
ICA + negentropy (blind source separation)	Model-free recovery of the aging component from (S, T, V) alone	Recovered component matches the regression-based correction at $ corr = 0.834$, using none of the measured T, V coupling — independent corroboration
Surrogate/null tests, n_{eff}, block-bootstrap	Statistical honesty on a short, autocorrelated series	Measured MI is 12× the estimator’s own bias floor — proven real, not artifact; every CI and credible interval uses $n_{eff} = 1,224$, not the raw 768,884 samples

Deliberately **out of scope**: concrete source-coding codecs (Huffman/LZW), channel error-correcting codes, and Big Data/streaming tooling — judged to correlate weakly with this specific research question and redundant with results already computed.

3. How it differs from and improves on each prior paper

Auto_Tuning_Aging_Sensor_... (Nogueira et al. — the sensor itself)

That paper **builds and validates the on-chip slack sensor**: it shows the sensor tracks critical-path delay degradation under burn-in, temperature, and voltage stress, at 19.85 ps per slack count on an Artix-7 FPGA. It is a hardware/validation contribution.

This work does not touch the sensor. It takes the sensor's output as a given noisy signal and asks a question that paper never poses: *how much information does this reading actually carry, and how much of it is recoverable noise?* It is a purely analytical layer sitting on top of that sensor's data.

Quantifying_the_Effect_of_Burn_In_Thermal_Control_... (Alencar et al. — thermal-fidelity study)

This is the paper **most directly superseded on its central metric**. It compares a bang-bang oven against a closed-loop PID oven on the same device and sensor, using **linear** R^2 (92.1% bang-bang vs. 59.2% PID) and residual standard deviation ($\sigma = 0.90$ vs. 0.73 counts) as its figures of merit for “how clean is this validation environment.”

This work shows, formally, why that comparison is incomplete: $\text{Cov}(X, Y) = 0$ does not imply independence, so a regression R^2 can read as “mostly explained” a variable that is, in fact, non-linearly and near-fully determined by its inputs — exactly the case here, since BTI aging is super-linear in V_{core} and Arrhenius acceleration is exponential in $1/T$. Mutual information is built from the full joint distribution rather than a second-moment summary, so it does not have this blind spot, and it is measured in a **transferable unit (bits)** rather than a variance fraction. The measured MI exceeds the prior paper's Gaussian-equivalent MI by 26% — a concrete, quantified version of “you were undercounting the coupling.” On top of that, this work adds three figures of merit the thermal-fidelity paper never had: a capacity/resolution estimate, a BSC alarm-reliability model, and a Cramér-Rao/Bayesian RUL detectability bound.

Sistema Automatizado de Monitoramento... (Alencar — TCC / burn-in platform thesis)

That thesis is the **data-generation infrastructure**: the PID-controlled, multi-threaded acquisition platform that logs slack, temperature, and voltage synchronized at 1 Hz. Its contribution is the instrument, described and validated on its own engineering terms (control loop, logging reliability, automation).

This work consumes that platform's output as raw material for an entirely different kind of analysis — it never modifies or re-validates the platform itself. Where the TCC treats the logged data descriptively (as a record that the automation worked), this work treats the same kind of log as the empirical substrate for entropy, mutual information, capacity, and RUL-detectability estimation — a new analytical use of data the TCC only needed to demonstrate acquisition worked.

4. One-line summary

The three prior papers **built the sensor, validated its thermal fidelity with a linear metric**, and **built the platform that generates its data**. This work adds the **measurement theory** on top of all three: turning “variance explained” into bits, and giving an SLM designer a resolution limit, an alarm-reliability curve, and an honest RUL detectability bound that none of the three could state.